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A palaeomagnetic study of hydrothermal activity and uranium mineralization at Mt Painter, South Australia

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Palaeomagnetic measurements indicate that the uraniferous Radium Ridge Breccias near Mt Painter in the northern Flinders Ranges of South Australia have been magnetized twice, both times in the Permo-Carboniferous. The palaeomagnetic south pole for one remanence was estimated as 165.1°E, 65.7°S (A_{95} = 11.5°). The direction for the other remanence was not well defined; however samples collected from diamictite bodies within the breccia gave 133.9°E, 33.1°S (A_{95} = 6.9°) as the probable pole for the overprint. The latter is similar to overprint poles published from central Australia, which are generally attributed to the Alice Springs Orogeny.

Magnetization directions interpreted as Permo-Carboniferous were obtained also from the Mt Gee Sinter (a quartz–hematite-rich chemical sedimentary rock which overlies the Radium Ridge Breccias) and from U-mineralized hematitic ironstone bodies within the breccia. The magnetizations of both units were probably coeval with the younger magnetization of the Radium Ridge Breccias. A positive fold test demonstrates that the remanence of the sinter is primary, indicating a major hydrothermal event in the Permo-Carboniferous.

It is not clear by how much this event post-dates the deposition of the underlying Radium Ridge Breccias and hematitic ironstone. The latter may be Ordovician or older if earlier monazite U–Pb data are correct, and may have been formed by granite-related hydrothermal fluids. In that case, the older remanence in the breccia is also an overprint. Epithermal sinter formation and chemical (rather than purely thermal) resetting of the remanence in the underlying breccia were probably due to deep circulation of oxidized fluids during Permo-Carboniferous tectonic activity. Chemical reaction of these fluids with pre-existing magnetite-bearing ironstone may have been responsible for uranium mineralization during the Permo-Carboniferous. This interpretation is consistent with published textural, isotopic and fluid inclusion data, but an older age for the uranium concentration, as a primary part of the ironstone formation, cannot be excluded. Alternatively, if the monazite data are discarded, the entire hydrothermal process including iron, uranium and silica deposition could have occurred in the Permo-Carboniferous.

Key words: hydrothermal, palaeomagnetism, sinter, uranium ore.

INTRODUCTION

The Mt Painter area, 550 km north of Adelaide in the northern Flinders Ranges of South Australia, is one of the world's earliest radium mining districts (Brown 1911). It has been explored extensively for uranium at various times and mapped in detail (Youles 1975, 1984, 1986; Major 1976; Drexel 1980, 1981; Drexel & Major 1987, 1990), but no major uranium mines have ever been established. The area contains a large number of minor copper, gold and other base metal occurrences, some of which were mined in the past. Recent exploration interest has been based partly on the recognition of siliceous sinter in the area (Drexel & Major 1987; Higgins *et al.* 1988), and partly on the striking similarity of some of the U-mineralized hematitic breccia at Mt Painter with breccia hosting the giant Proterozoic Cu–U–Au–Ag–REE deposit at Olympic Dam (Youles 1978, 1984; Roberts & Hudson 1983; Oreskes & Einaudi 1990).

The Mt Painter Inlier exposes Middle Proterozoic crystalline basement and is surrounded by Adelaidean (Late Proterozoic) sedimentary sequences. The region has experienced a complex thermal history extending through to present-day hot spring activity. The Phanerozoic thermal history of the inlier was probably influenced by the abundance of unusually uranium-rich highly fractionated alkali-feldspar granite, regarded as Palaeozoic in age (Teale 1979; Teale & Lottermoser 1987; Drexel & Major 1990). The breccia is composed dominantly of locally derived basement clasts which are altered and cemented to varying degrees by K-feldspar, quartz and hematite. Breccia is distributed over an area of at least 8 × 10 km near the centre of the inlier (Fig. 1), and comprises forms varying from massive breccia piles to funnel-shaped bodies, with water-laid and subaerial chemical and mechanical sedimentary strata which are still in a near-horizontal bedding position (Youles 1975; Major 1976; Drexel & Major 1987, 1990; Higgins *et al.* 1988). The origin of the breccia and contained ores is debated; interpretations ranging from magmatic-hydrothermal activity associated with Ordovician (or younger) granitoid intrusion (Drexel 1980; Lambert *et al.* 1982), through deep hydrothermal fluid convection driven by the radiogenic heat of high heat-producing granite (Higgins *et al.* 1988), to low-temperature supergene processes associated with deep weathering in the Tertiary (Blissett *in* Coats & Blissett 1971).

Any information on the absolute timing of breccia formation and mineralization impacts directly on these widely disparate genetic interpretations. The typically

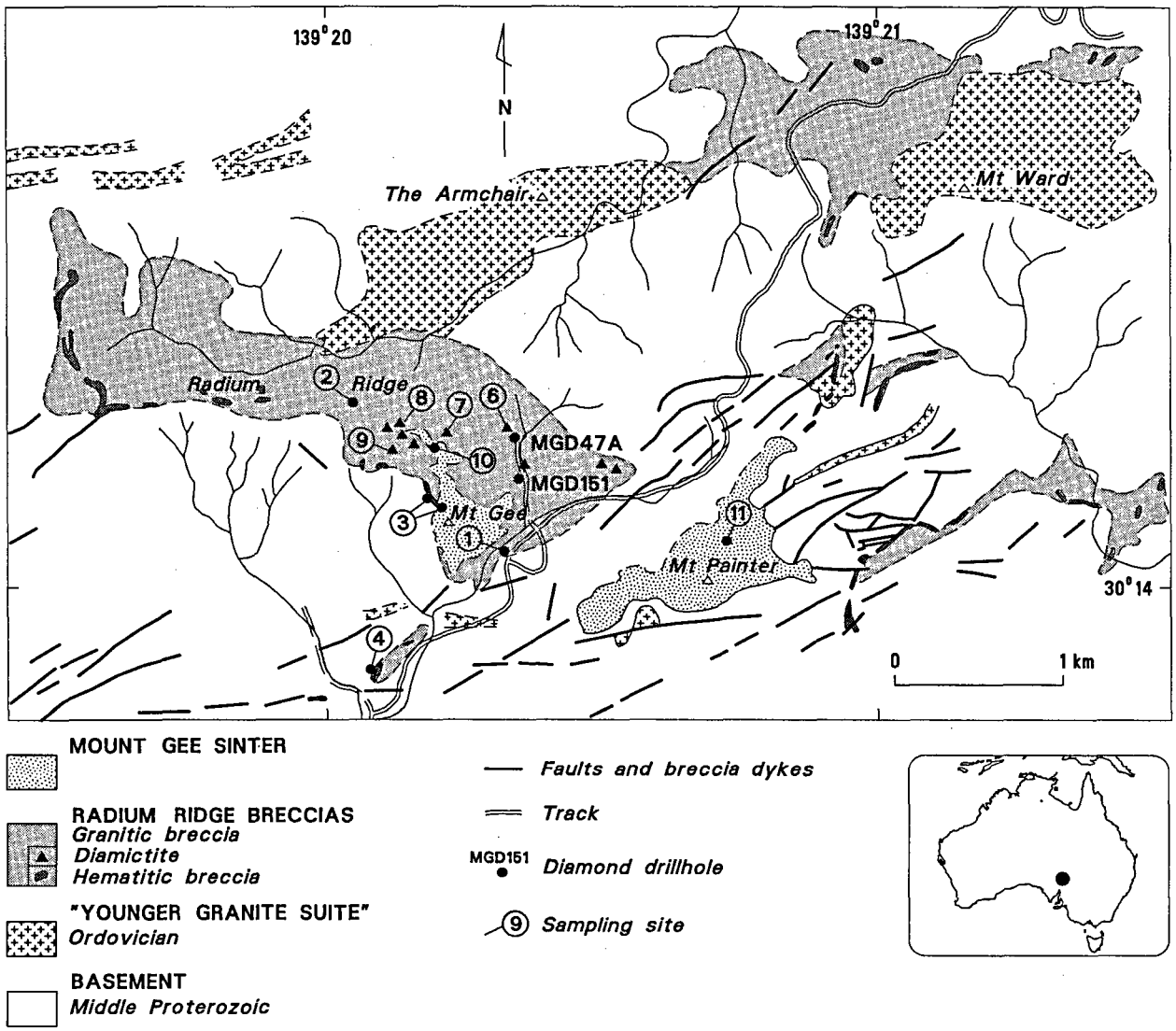


Fig. 1 Geologic map with sampling localities of the Mt Painter area (after Drexel & Major 1987). Locality 5, at Hematite Valley 8 km northeast of Mt Painter, is outside the map area.

high magnetic stability of hematite, the ubiquitous iron mineral in all breccia types at Mt Painter, makes palaeomagnetism a potentially powerful tool to date the primary age of deposition of the sedimentary-hydrothermal complex by comparison with the Australian polar wander path, and may provide some additional information about the hydrothermal history of the breccia after its deposition. The present paper reports the results of a case study to test this approach to dating hydrothermal ore-forming events, so far used only in a few other studies (e.g. Reynolds *et al.* 1985; Browning & Beske-Diehl 1986; Jowett *et al.* 1987; Pan *et al.* 1990; Symons & Sangster 1991).

GEOLOGY

Regional geological history

The geology and mineralization of the region have been described and mapped by Sprigg (1945), Coats and Blissett (1971), Teale (1979), and other workers whose results have been compiled by Preiss (1987). These studies provide the basis for the following summary.

The Mt Painter Inlier exposes roughly equal areas of granitoid (of various intrusion ages) and metamorphosed supracrustal rocks. The oldest polymetamorphic basement in the inlier includes gneiss, metavolcanic rocks and schist of sillimanite-grade, intruded by an early suite of granite (Fig. 1). Based on lithological similarities, these rocks have been correlated with the high-grade metamorphic rocks of the Middle Proterozoic Willyama Supergroup hosting the Broken Hill lead–zinc deposit (280 km southeast of Mt Painter), but no geochronological studies have been carried out at Mt Painter to test this correlation.

The high-grade crystalline rocks and some possibly younger less-metamorphosed and deformed quartzite, phyllite and metavolcanic rocks (G. S. Teale, quoted by Preiss 1987, p. 38) are unconformably overlain by a thick sequence of Adelaidean (Late Proterozoic) rocks included

in the Adelaide Geosyncline. The latter comprise dominantly shallow-marine clastic, pelitic and evaporitic carbonate and minor mafic volcanic rocks. Cambrian limestone, sandstone and siltstone overlie the Adelaidean sequence. The Delamerian Orogeny in the Cambro-Ordovician caused open folding throughout most of the present Flinders Ranges. A Delamerian high-grade metamorphic overprint is restricted to the Mt Painter Inlier and to a narrow halo (<5km) in the surrounding Adelaidean sequence, where low-pressure high-temperature metamorphic assemblages including andalusite, cordierite and abundant scapolite were formed (Coats & Blissett 1971; Richert 1976). A younger suite of granitoid rocks (Cambro-Ordovician on the basis of Rb–Sr dating; Compston *et al.* 1966), including highly fractionated and uranium-rich alkali-feldspar muscovite granite, tourmaline albitite and pegmatite, intruded the basement and cut locally across the Adelaidean metasedimentary sequence (Fig. 1). All basement rocks at Mt Painter show unusually high uranium contents averaging 30 ppm (Drexel & Major 1990, table 1).

Except for the enigmatic breccia units, no stratigraphic record is preserved in the Mt Painter area between the time of intrusion of the Cambro-Ordovician granitoid bodies and the Tertiary, when major north-trending faults established the main topographic features of the present Flinders Ranges and created the surrounding Lake Frome, Lake Eyre and Lake Torrens basins. These basins are filled with shallow-marine to lacustrine carbonate, fluviatile clastic and evaporite deposits (Callen 1977). The Paralana–Lady Buxton Fault system, along which the Mt Painter Inlier was uplifted in the Late Tertiary or Pleistocene, locates some hot springs discharging radon-rich waters with temperatures above 60°C (Coats & Blissett 1971). Drainage from breccia-hosted uranium mineralization in the Mt Painter region was probably the main contributor to second-stage roll-front uranium deposition in Tertiary sedimentary deposits in the Lake Frome basin to the east (e.g. Beverley deposit; Haynes 1975).

Uraniferous breccia in the Mt Painter region

The breccia deposits and associated uranium mineralization in the Mt Painter and Mt Gee areas have been described by Stillwell and Edwards (1954), Youles (1975, 1984, 1986), Major (1976), Drexel (1980; 1981), Lambert *et al.* (1982), Drexel and Major (1987, 1990) and Higgins *et al.* (1988). The breccia comprises a pile of variable thickness up to at least 500 m (Drexel & Major 1987), grading from *in situ* fractured and K-feldspar-altered basement to dominantly granitic and hematitic breccia units (the Radium Ridge Breccias) and finally to vuggy rocks consisting dominantly of quartz–hematite-rich chemical precipitates near the top (the Mt Gee Sinter; Drexel & Major 1987; Fig. 2a). The breccia is restricted to the centre of the Mt Painter basement block and there is no evidence of stratigraphic relation to the Adelaidean sequence. It shows no metamorphic recrystallization or deformation that could suggest a Delamerian regional-metamorphic overprint. No potentially coeval volcanic rocks have been identified in the region (Drexel 1980).

Granitic breccia

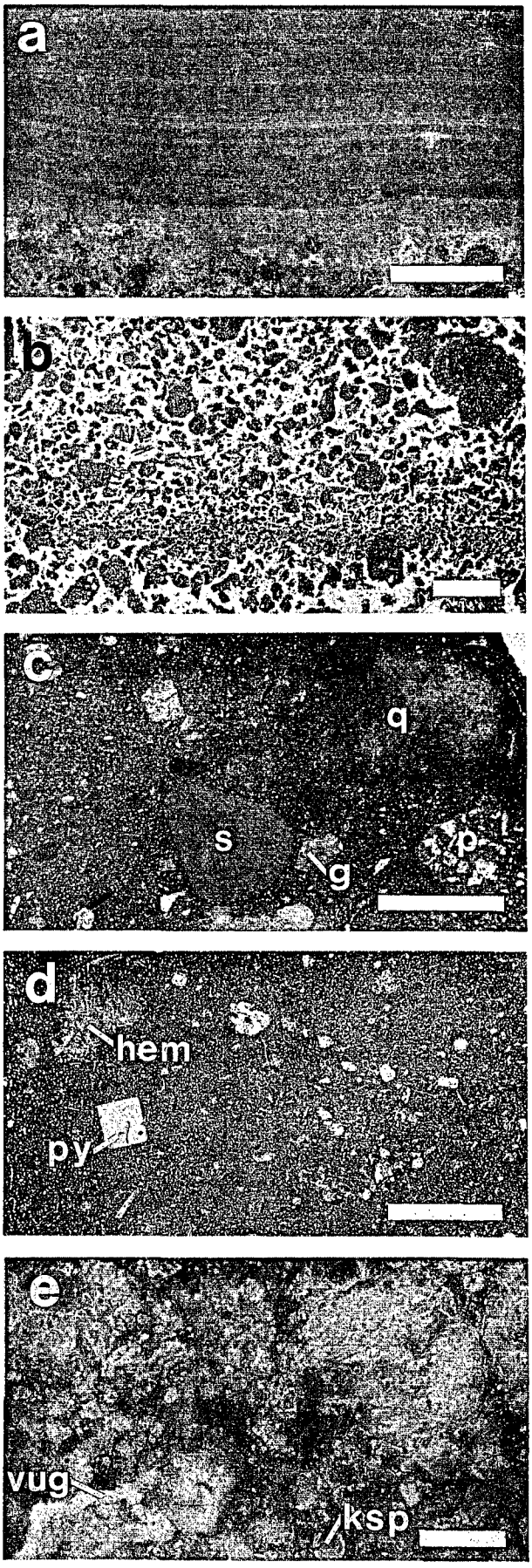
Granitic breccia, the dominant breccia type, comprises mainly clasts of local basement granite, gneiss and schist, ranging from house-sized blocks to silt-sized fragments. Sorting and rounding of clasts is minimal, and well-defined bedding is restricted to rare, discontinuous lenses of red siltstone and arkose. Partial or pervasive K-feldspar alteration of basement clasts is widespread (Lambert *et al.* 1982; Fig. 2e). Veins and vugs lined with euhedral quartz crystals similar to those in the overlying Mt Gee Sinter (Fig. 2e) are common.

Hematitic breccia

Hematitic breccia is host to most of the uranium mineralization in the region. It occurs as lens-shaped bodies, up to 70 m thick, at the base and within the granitic breccia. Contacts range from subhorizontal and apparently layer-parallel, to subvertical and apparently cross-cutting (intrusive into?) the granitic breccia. With local exceptions at the metre scale, bedding is absent in the hematitic breccia. The most typical lithology consists of a chaotic mixture of partly broken and reprecipitated specularite laths, ochreous hematite, quartz and minor granitic fragments in variable proportions, plus a large number of minor minerals (Fig. 2d). These include, in order of paragenetic sequence: monazite, pyrite, chalcopyrite, molybdenite, magnetite, specular hematite, clear quartz, uraninite, later Cu–Fe–Pb–Zn–sulphide minerals including chalcocite and bornite, followed by carbonate, chlorite, fluorite and later cloudy quartz. Some of the hematite pseudomorphs earlier magnetite (Stillwell & Edwards 1954; Drexel & Major 1990).

Diamictite

Diamictite occurs in small outcrops and drillhole intersections along a distinct east–west trend (Fig. 1). In sharp contrast to the enclosing granitic breccia, the diamictite contains a variety of well-rounded clasts, including porphyritic volcanic and other rocks that do not occur in the local basement (Fig. 2c). The cobbles are supported by a silty to sandy matrix which is undeformed and unrecrystallized. The matrix is generally red or brown, except for parts of outcrop 9 (Fig. 1) where green portions have been partly overprinted(?) along sharp redox fronts by red material. Where exposed in three dimensions, the diamictite bodies show sharp and steep contacts with the granitic breccia, but no contact relations to hematitic breccia or quartz–hematite sinter were observed. The diamictite has been interpreted variably as stratigraphic intercalations of Adelaidean glaciomarine strata (on the basis of their striking lithological similarity with massive Sturtian tillite adjacent to the inlier; Youles 1975, 1986); as solid rafts that were introduced tectonically (Drexel & Major 1987, 1990); as pebble-breccia dykes (Drexel & Major 1987); and as soft-sediment intrusive 'dykes' (Sprigg 1989).



Mt Gee Sinter

Mt Gee Sinter, an irregularly layered unit up to 200 m thick, is composed almost exclusively of hydrothermal precipitates of quartz, hematite and minor zeolite (laumontite and stilbite; mainly represented by negative-crystal pseudomorphs; Stillwell & Edwards 1954). The sinter (Fig. 1) consists of a mixture of three main rock types: massive or vuggy hydrothermal quartz with abundant metre-sized crystal-lined cavities; layers consisting of white hydrothermal quartz that apparently supports centimetre-sized clasts of coarse specular hematite, with distinct 'graded bedding' in places (quartz–specularite sinter; Fig. 2b); and finely laminated reddish-brown chemical sedimentary strata of silt-sized quartz with minor fine-grained hematite and submicroscopic, presumably hematitic, pigment (here referred to as fine-grained siliceous sinter; Fig. 2a). The overall dip of the Mt Gee Sinter is shallow, but steep dips occur locally near irregular metre scale dome-and-basin structures, which are marked mainly by the laminated fine siliceous sinter. This deformation must have occurred while the laminated material, now extremely hard and chert-like, was still plastic (Drexel & Major 1987) and is probably related to local adjustment due to differential compaction of the underlying vuggy sinter. These unusual rocks could have been originally deposited as silica-rich gel or sodium silicate precursors such as magadiite, as described by Eugster and Jones (1967) and Eugster (1969). This would imply relatively alkaline hot-spring solutions, consistent with the presence of zeolite minerals in the sinter, the predominance of alkali feldspar and the rarity of clay minerals in the underlying breccia.

Previous age constraints on breccia formation and mineralization

Youles (1975) and Major (1976) inferred a Sturtian (~650 Ma) age for the breccia complex on the basis of the striking lithological similarity of the diamictite at Mt Painter with glaciomarine strata in the nearby Umberatana Group. However, cross-cutting (intrusive?) relationships at improved exposures now suggest that the diamictite is probably not part of the sedimentary breccia stratigraphy (locality 6, Fig. 1; R. C. Sprigg pers. comm.

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Fig. 2 Close-up photographs of slabs of typical rock types of the Mt Gee Sinter (a, b) and the Radium Ridge Breccias (c, d, e). Scale bars = 10 mm. (a) Fine-grained siliceous sinter (laminated) overlying coarse quartz-specularite sinter. (b) Coarse quartz-specularite sinter with 'graded bedding'. (c) Diamictite with round pebbles of exotic porphyry (p), granite (g), quartzite (q) and siltstone (s). (d) Hematitic breccia with euhedral pyrite (py) and clasts of specularite (hem). (e) Granitic breccia containing some clasts rimmed by K-feldspar (ksp), and a vug lined with quartz crystals.

1988). The interpretation of a Sturtian age has also been questioned because of the absence of a Delamerian metamorphic overprint on the breccia and diamictite (Drexel 1980; Lambert *et al.* 1982; Drexel & Major 1987).

The hematitic breccia post-dates Delamerian magmatism and subsequent erosion because it contains clasts of pegmatite that are very similar to a post-Adelaidean pegmatite which has been dated by Rb/Sr at ~460 Ma (Compston *et al.* 1966). Three monazite samples from the hematitic breccia have been dated by the U–Pb method (Pidgeon 1979). They define a poor discordia line interpreted by Pidgeon (1979) in terms of a Delamerian primary age of 440 ± 50 Ma and minor recent lead loss. However, Youles (1986) pointed out that the results could also be interpreted, within analytical error, to represent an older primary age of 650 ± 100 Ma (i.e. Sturtian) with a major event of lead loss at 280 ± 40 Ma. Youles (1986) placed no major significance on this reset age which, interestingly, corresponds to the Early Permian when a major remagnetization event is recorded (see below).

SAMPLING AND LABORATORY TECHNIQUES

For the palaeomagnetic study, more than 200 samples were collected from granitic and hematitic breccia units, sinter deposits and diamictite at eleven localities in the Mt Painter region (Fig. 1). Samples from the sinter were collected as blocks; and those from the other rock units as cores, the latter by means of a portable diamond drill. Sample orientations were obtained with a magnetic compass and, nearly always, a sun compass.

The drillcores were cleaned of surface contaminants by sandblasting with fine glass grit. Strongly magnetized specimens (>200 mA/m) were measured either on a Digico spinner magnetometer or a three-axis 2G cryogenic magnetometer. Weaker specimens were usually measured on a two-axis ScT cryogenic magnetometer. The specimens were demagnetized thermally in either a small Schonstedt TSD-1 demagnetizer or large furnaces described by McElhinny *et al.* (1971). Magnetic susceptibilities were measured between heatings to monitor chemical changes. Before thermal treatment, some samples were partly demagnetized in alternating fields (AF) using a Schonstedt 2-axis GSD-5 tumbler-demagnetizer. Except where the demagnetization was terminated prematurely because of large directional scatter, the minimum number of demagnetization steps was 18. For many samples the final demagnetization stage was very abrupt, so that ten or more additional steps were required at 1°C nominal spacing. The data acquisition and processing system described by Giddings (1985) was used. The various components of remanence were determined from the demagnetization data by principal component analysis (Kirschvink, 1980; adapted by C. T. Klootwijk, Australian Geological Survey Organisation). Unless noted otherwise, only directions with a mean angular deviation of 10° or less were used to calculate palaeomagnetic poles.

RESULTS

Granitic breccia

Preliminary work on the granitic breccia gave poorly defined magnetic directions, and subsequent sampling was focused on small arkosic lenses within the breccia. Unlike their host, most of these fine grained lenses have a distinct layering which is assumed to have been originally subhorizontal (Drexel & Major 1987, p. 16). This assumption is consistent with a fold test between two different granite breccia outcrops (below), although soft-sediment deformation may have occurred locally as well.

Twenty-three samples were collected from two localities, one immediately south of Mt Gee, the other on the southern slopes of Radium Ridge (localities 1 and 2 on Fig. 1). The first locality was adjacent to a fault zone and layering had a substantial dip; the second was well away from mapped faults and had sub-horizontal layering.

Remanence intensities were initially in the range 3–270 mA/m and remained nearly constant during heating steps until the temperature reached close to the Curie Point of hematite (Fig. 3a), which suggests a high magnetic stability. Three remanence components were found, characterized by low, intermediate and high unblocking temperatures. The last two are seen as linear segments in the orthogonal projection plots of Fig. 3a. Removal of the final residue of the intermediate component was often marked in the granitic breccia and some of the other samples by a small inflection in the normalized intensity plots (Fig. 3a). No corresponding change was observed in magnetic susceptibility that could indicate a change in magnetic mineralogy during heating.

The low unblocking temperature component had a negative inclination and appears to be a viscous magnetization acquired in recent fields. The other two components had positive inclinations. A component with intermediate unblocking temperature was generally seen in the orthogonal projection plots as a short linear segment comprising scattered points (Fig. 3a); it was resolved with mean angular deviation (MAD) < 20° in only eight samples, and yielded a loose group of directions on the stereo diagram (see Fig. 4a). The grouping appears to be marginally more scattered when bedding corrections are applied, but the difference is not statistically significant. The mean pole, corrected for bedding, is 151.9°E, 53.7°S (A_{95} = 24.9°; see footnote Table 1).

The high unblocking temperature component, which dominates the remanence, was well defined in all specimens (MAD < 5°). Directional scatter between samples was high before bedding corrections, but improved markedly after the corrections were made (Fig. 4a). The fold test of McFadden and Jones (1981) confirms that the improvement is statistically significant; however, the time of fault-related deformation is unknown so that the fold test has limited value in constraining the age of magnetization. The mean pole is 165.1°E, 65.7°S (A_{95} = 11.5°).

The results from the granitic breccia together with those from the other units are summarized in Table 1.

Table 1 Summary of Palaeomagnetic Results.

Rock unit/ Locality	Component	Location Co-ordinates		No. of samples	Remanence direction								Pole			
					Before tilt corrections				After tilt corrections							
		Longitude (°E)	Latitude (°S)	(specimens)	Declin. (deg)	Inclin. (deg)	k	α_{95} (deg)	Declin. (deg)	Inclin. (deg)	k	α_{95} (deg)	Longitude (°E)	Latitude (°S)	K	A_{95} (deg)
Granitic breccia																
Locality 1	H	139.34	30.23	10(20)	199.8	34.7	29.171	9.1	164.5	66.7	29.224	9.1	163.7	66.2	3.025	13.9
Locality 2	H	139.33	30.22	10(17)	158.3	67.4	14.621	13.1	161.8	65.6	13.232	13.8	166.5	65.2	6.363	20.8
Overall average	H	139.34	30.23	20(37)	187.1	52.4	9.089	11.5	163.1	66.2	19.192	7.7	165.1	65.7	9.017	11.5
Overall average	I	139.34	30.23	8(9)	189.9	66.6	16.875	13.9	162.2	75.1	14.041	15.3	151.9	53.7	5.904	24.9
Hematitic breccia																
Locality 3	H	139.34	30.23	8(11)	203.3	75.9	5.730	25.4					125.7	50.9	2.727	41.6
Locality 4	H	139.33	30.24	8(8)	47.6	84.3	4.151	31.0					146.3	26.1	2.260	48.5
Locality 5		139.42	30.17						Inconsistent results							
Overall average	H	139.34	30.23	16(19)	189.7	85.1	4.844	18.8					137.3	39.6	2.500	29.9
Diamictite																
All localities	H	139.34	30.23						Random directions							
Locality 6	I	139.34	30.22	21(32)	227.5	81.7	31.886	5.7	—	—	—	—	125.8	39.7	11.129	10.0
Locality 7	I	139.34	30.23	10(13)	170.9	86.8	22.683	10.4	—	—	—	—	141.3	37.5	7.793	18.5
Locality 8	I	139.34	30.22	5(6)	39.7	71.1	6.623	32.1	—	—	—	—	154.2	8.7	4.075	41.7
Locality 9	I	139.34	30.23	26(37)	284.0	86.5	24.114	5.9	—	—	—	—	133.2	29.6	8.205	10.6
Overall average	I	139.34	30.23	62(81)	248.2	87.2	19.636	4.2	—	—	—	—	133.9	33.1	7.753	6.9
Overall average	L	139.34	30.23	45(59)	10.0	−61.2	9.686	7.2	—	—	—	—	119.4	73.3	8.489	7.8
Fine-grained siliceous sinter																
Locality 10	H	139.34	30.23	13(24)	160.3	54.6	5.293	19.9	137.9	77.6	12.398	12.3	160.4	44.6	4.630	21.7
Locality 11	H	139.35	30.23						Large directional scatter							
Locality 10	I	139.34	30.23	14(30)	154.0	61.5	5.072	19.6	110.0	81.5	13.444	11.3	157.8	35.3	4.890	20.1

k, K are the precision parameters of Fisher's distribution; α_{95}, A_{95} are the semiangles of cones of 95% confidence.
Component: H = high unblocking temperature; I = intermediate unblocking temperature; L = low unblocking temperature.

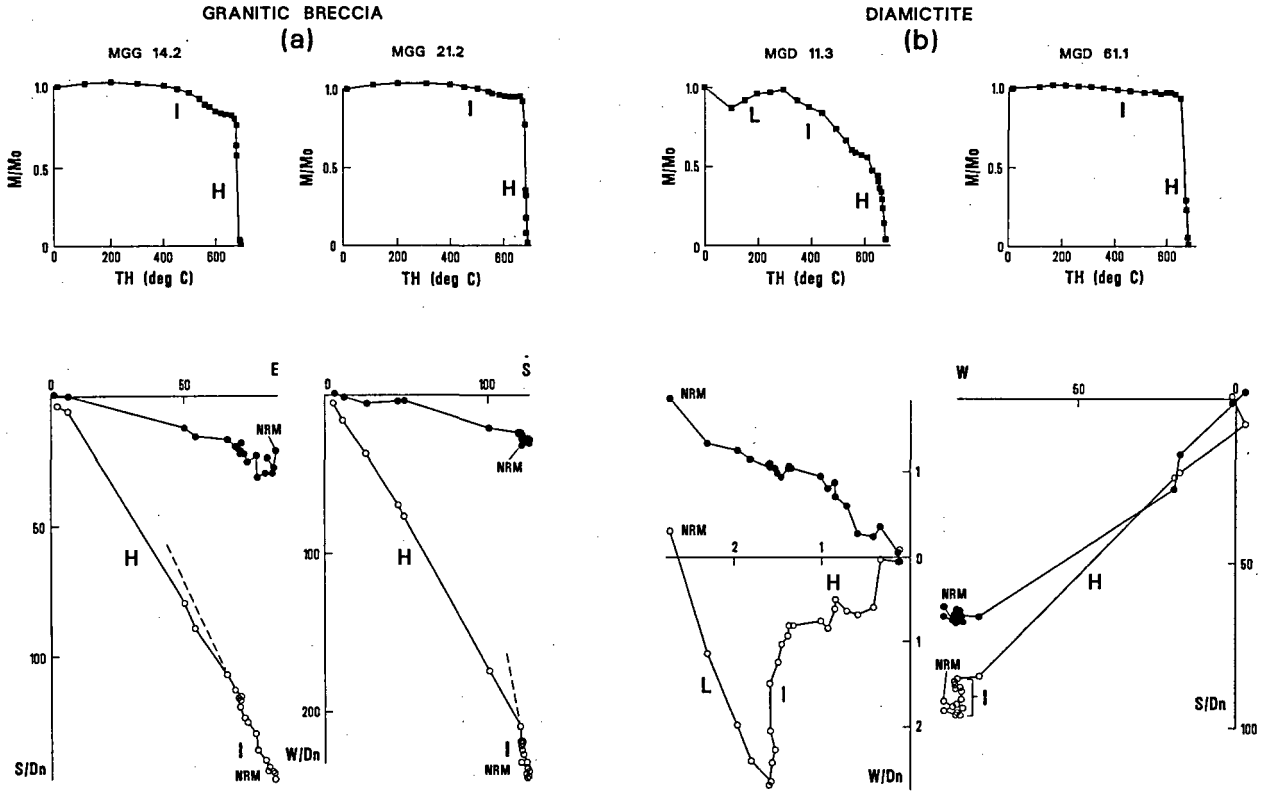


Fig. 3 Representative thermal (TH) demagnetization plots for (a) granitic breccia and (b) diamictite. Top diagrams are normalized intensity plots. Bottom diagrams are orthogonal projection plots of the magnetization vector end points. Open symbols indicate projection onto a horizontal plane; closed symbols indicate projection onto a vertical plane (Dn = down). The axial units are mA/m. The orthogonal projection plot in (a) contains bedding-corrected directions, the plot in (b) contains uncorrected directions. Letters L, I and H indicate low, intermediate and high unblocking temperature components, respectively. Dashed lines in (a) indicate intermediate component trend. In some samples like MGG 14.2 the I and H component directions are indistinguishable and the components can only be identified in the normalized intensity plots (small inflection near 600°C).

Diamictite

Eighty-two samples were collected from four localities near Mt Gee (6–9 in Fig. 1). Twelve were from single cobble-sized clasts in the diamictite, and included a variety of clastic sedimentary rocks, schist, granite and porphyry.

Initial intensities of remanence ranged from 0.5 to 750 mA/m, the large spread reflecting the inhomogeneity of the rock. Like the granitic breccia, this unit contained three components of remanence (Fig. 3b). The first had a low unblocking temperature and negative inclination; it was removed typically in the temperature range 100–250°C and yielded the mean direction 10.0°, -61.2° (A_{95} = 7.2°) which is close to the present field direction at the site (7.3°, -62.5°).

The second component was characterized by an intermediate unblocking temperature and positive inclinations. It was found in both single cobbles and composite samples. Even though bedding was not discernible in the diamictite, other evidence indicates that these directions require no tilt correction: except near faults and at sites of plastic deformation of the sinter, tilts at the Mt Gee sampling region appear to be negligible. The layering in two drillholes immediately northeast of Mt Gee (MGD47A and MGD151 in Fig. 1) is near horizontal (Drexel & Major 1987), as is the layering of arkose lenses to the west of Mt Gee (at the second granitic breccia site discussed above) and that of sinter terraces on the north ridge of Mt Painter (below). Drillhole MGD47A is located next to one of the diamictite sites. The directions for the intermediate component are shown in Fig. 5a; its mean pole is 133.9°E, 33.1°S (A_{95} = 6.9°).

The third component had a high unblocking temperature and a magnitude that varied widely from sample to sample: three of the single-cobble samples contained no trace of this component, presumably because of the dominance of the intermediate component. In most samples, the directions were poorly defined; they are shown in Fig. 5b, where the mean angular deviation acceptance criterion has been relaxed from $< 10°$ to $< 20°$ and where specimen, rather than mean sample, directions have been plotted. The plot itself is random.

Because of its moderate negative inclination and similarity to the present field direction, the low unblocking temperature component appears to be due to recent fields. The high unblocking temperature component, being random, must have been acquired before the cobbles were incorporated in the diamictite. The origin of the intermediate temperature component is discussed in the next section.

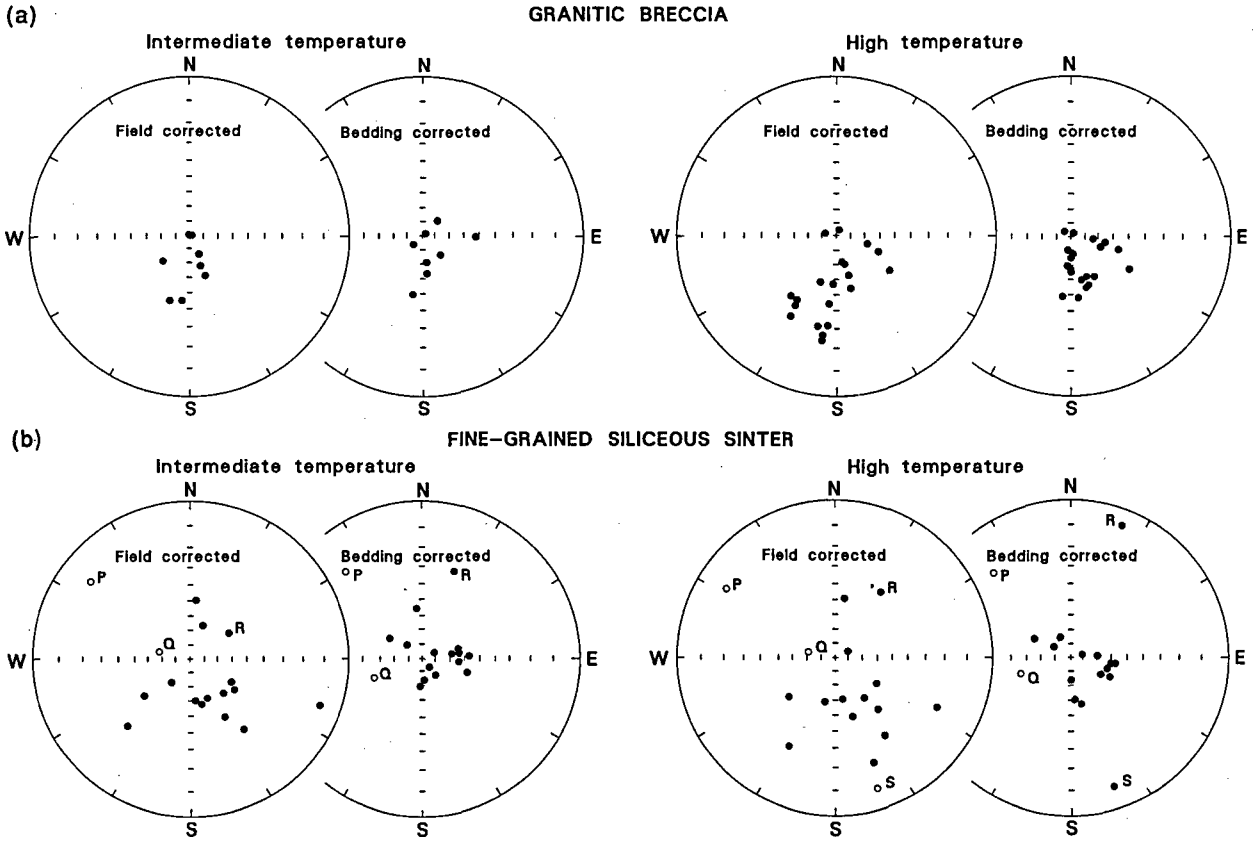


Fig. 4 Equal angle stereo projections showing intermediate and high unblocking temperature directions for (a) granitic breccia and (b) fine-grained siliceous sinter. Open (closed) circles denote upper (lower) hemisphere. Intermediate component directions for the granitic breccia have mean angular deviation (MAD) $<20°$, otherwise MAD $<10°$. Directions P, Q, R, S in (b) are discussed in the text. Most samples contained an I component but in some like MGD 61.1 (Fig. 3b) and MGS 53.2 (Fig. 8a) it was represented by a short segment with poorly defined directions (MAD $>10°$) that were not used in the pole calculation

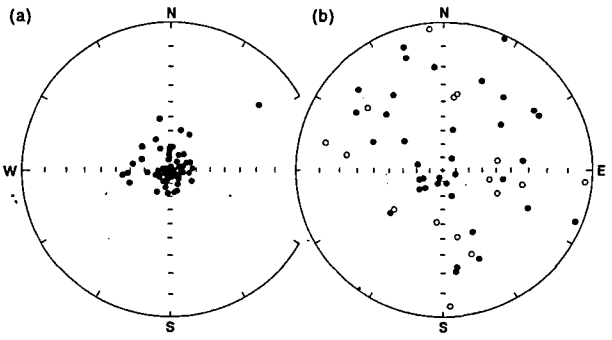


Fig. 5 Equal angle stereo projections of remanence directions in diamictite: (a) intermediate unblocking temperature component (MAD $<10°$), (b) high unblocking temperature component (MAD $<20°$). Symbols as for Fig. 4.

Hematitic breccia

Thirty-two samples were collected from two localities, the first near the summit of Mt Gee and the second at Painter No. 6 Uranium Workings (locality 4 on Fig. 1). Samples were also collected from arenite lenses within hematitic breccia at a third locality northeast of Mt Painter. The latter gave spurious directions which were discarded.

The NRM intensities ranged from 600 to 140 000 mA/m, although only few exceeded 5000 mA/m. Two demagnetization procedures were used. The first, standard thermal demagnetization, had little effect on remanence intensities until the temperatures reached close to the Curie Point of hematite (Fig. 6a). The resulting directions were highly dispersed, though not random (Fig. 7a). Since the sampling localities were high on hillslopes, we suspected that the large directional scatter is due to unresolved secondary components induced by lightning strikes. Such components have been found difficult to isolate by thermal demagnetization, whereas alternating field treatment has proved effective in some studies (Cox 1961; Graham 1961; Purucker 1974). Therefore, a second series of demagnetizations was carried out on the fewer remaining specimens by using alternating fields to remove low coercivity components, followed by thermal treatments. A large component was indeed removed at 20–100 mT peak fields, leaving a component with a high unblocking temperature (Fig. 6b, also Fig. 8a). There was also some evidence for a small intermediate temperature component (e.g. inflections in the normalized intensity plots at about 650°C). This component had a positive inclination, but was invariably too small to be determined

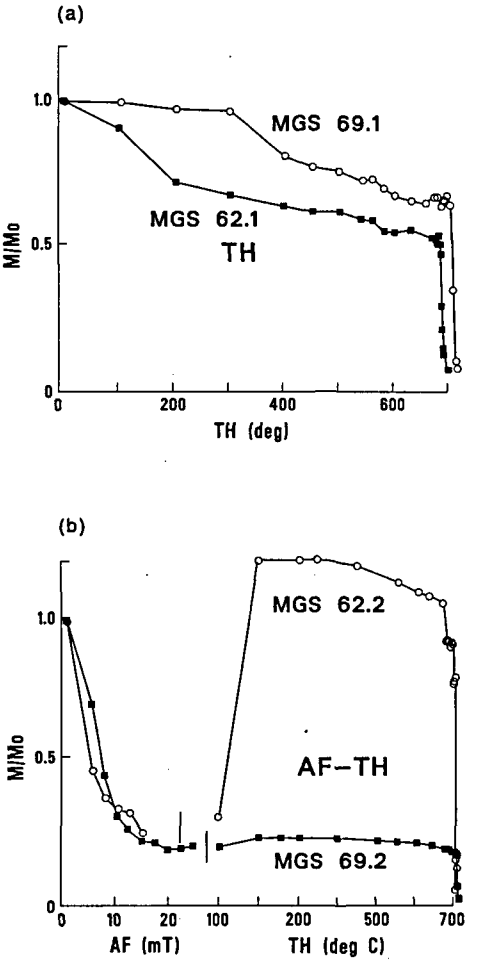


Fig. 6 Normalized intensity plots for two pairs of representative specimens of hematitic breccia, each pair being from the same sample: (a) specimens demagnetized thermally (TH), (b) specimens first partly demagnetized by alternating fields and then demagnetized thermally (AF–TH). A large remanence component is usually removed at relatively low peak fields, as seen especially in the plot for MGS 62.2 (in which the remanence direction became reversed during AF demagnetization).

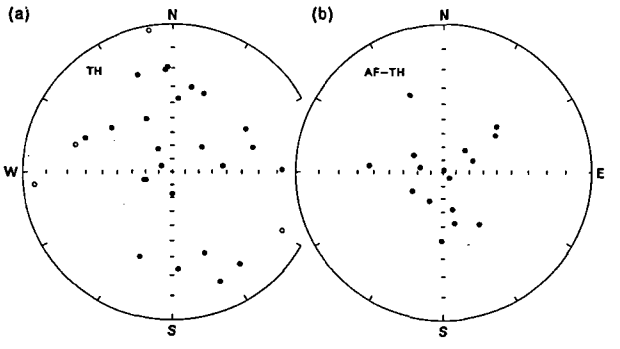


Fig. 7 Equal angle stereo projections for remanence directions in hematitic breccia: (a) directions obtained by thermal demagnetization only, (b) directions obtained by thermal demagnetization after AF treatments. All directions have been corrected for bedding. Symbols as in Fig. 4.

reliably. The directional scatter between samples for the high unblocking temperature component was considerably reduced by the combined alternating field–thermal treatment, though it remained appreciable (Fig. 7b). It is not clear if this scatter is due to a residue of the secondary component or some other effect (e.g. rotation during re-brecciation and/or chemical overprinting of the breccia). The mean pole is 137.3°E, 39.6°S ($A_{95} = 29.9°$).

Mt Gee Sinter

Two localities were sampled. The first is on the north ridge of Mt Painter, where fine-grained siliceous sinter crops out as several small sub-horizontal terraces. The remanence directions from these specimens were highly scattered. Alternating field demagnetization of pilot specimens from some of the most dispersed samples, followed by thermal demagnetization, did not alter the directions. The results from this locality were discarded.

The second locality is on the northern end of the Mt Gee ridge (locality 10 on Fig. 1) where coarse quartz–specularite sinter (Fig. 2b) is commonly capped by a few millimetres to several centimetres thickness of the fine-grained siliceous sinter (Fig. 2a). The latter is red and laminated, the laminae being defined by varying pigment concentrations and textural differences. The upper surface of the fine-grained siliceous sinter is parallel to the laminae and is usually deformed into metre-scale domes and basins with local dips up to 40°. Both the coarse- and fine-grained lithologies were sampled.

Samples from the quartz–specularite sinter had high initial intensities of remanence, ranging from 10 000 to 60 000 mA/m, except for one sample which was well above this range at 120 000 mA/m. Orthogonal projection plots indicated well-defined linear segments. Where a second specimen from the same sample was demagnetized, the agreement between specimens was usually good; however, the remanence directions between the samples differed widely (Fig. 9). It should be noted that layering orientations were not available for the quartz–specularite sinter, and the directions have, therefore, not been corrected for possible local tilts. This may explain at least part of the scatter (see below).

The initial remanence intensities of the fine-grained siliceous sinter were generally in the range 3–50 mA/m. Three components of magnetization were resolved, characterized respectively by low, intermediate and high unblocking temperatures (Fig. 8b). The last two components have similar directions. No break in slope in the orthogonal projection plots was observed for several specimens, but even for these the normalized intensity plots contained small inflections where the intermediate unblocking temperature component was removed in the other samples. Both components had a large directional scatter which was reduced markedly by rotating the laminae to a horizontal attitude about local strikes (Fig. 4b). In three samples, marked P, Q and R in Fig. 4b, the intermediate component directions departed considerably from those of other samples even after these rough corrections, two of the directions having negative inclinations.

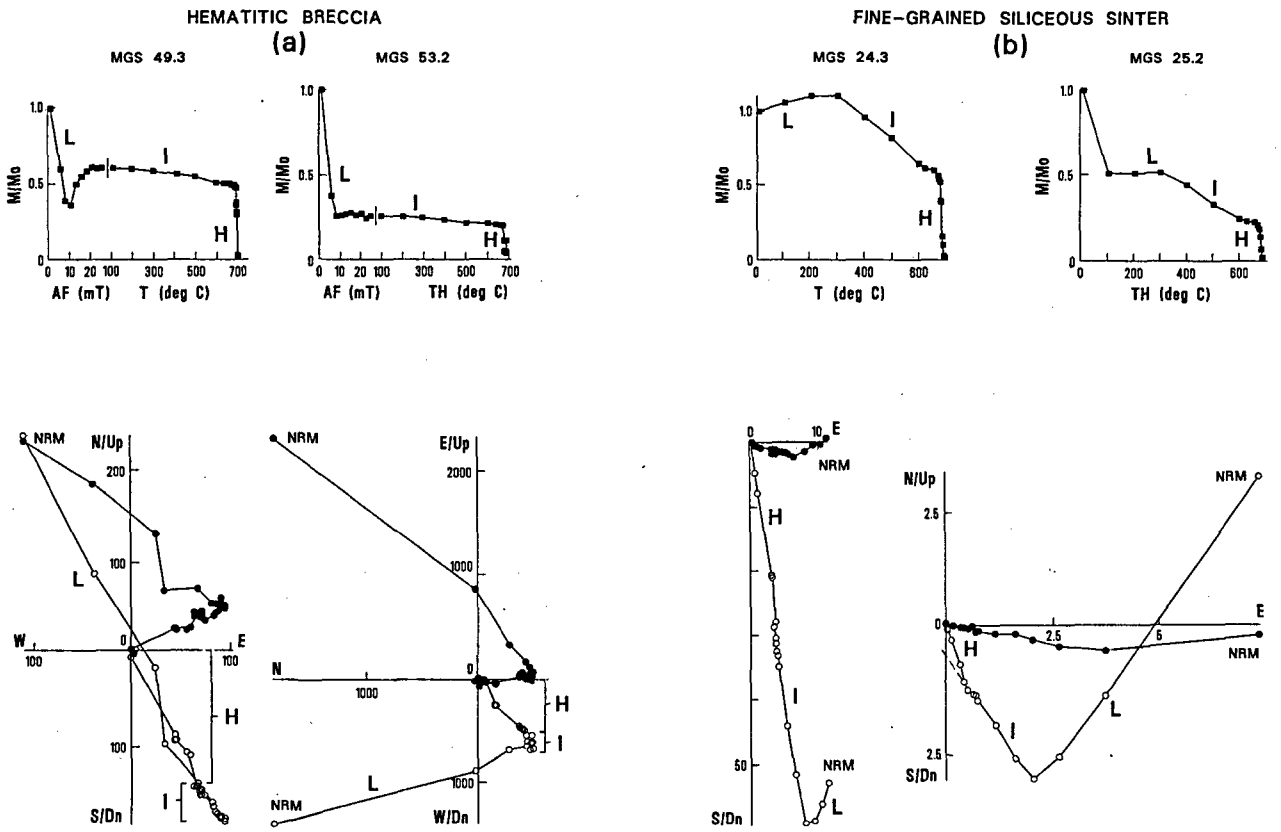


Fig. 8 Representative demagnetization plots for (a) hematitic breccia and (b) fine-grained siliceous sinter. Symbols as for Fig. 3. The magnetization vectors in (b) are bedding corrected, those in (a) are uncorrected. I and H components for MGS 24.3 are only slightly distinguished in the orthogonal projection plot, but are clearly separated by a small inflection in the normalized intensity plot.

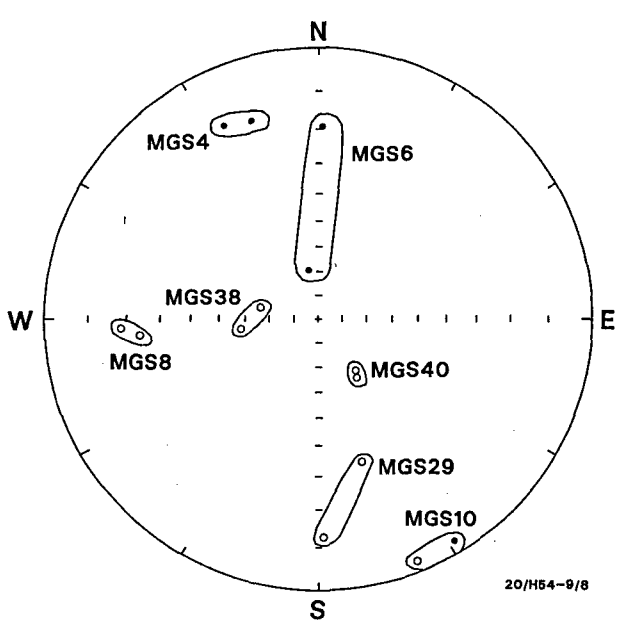


Fig. 9 Equal angle stereo projections for pairs of specimens from coarse quartz-specularite sinter, each pair being from the same sample. Note that the directions within most pairs agree within 10°, but the directions between pairs are widely scattered. Symbols as in Fig. 4.

The same three samples gave anomalous scatters for the hard component, as did a fourth sample (S) in which the intermediate component was not resolved. It is not clear what process has caused the discrepancies, but we noted a tendency for the scatter to increase after the laminae were rotated to horizontal. It, therefore, seems likely that the process is secondary (e.g. induced by lightning strikes). The four divergent directions were excluded from the analysis. The fold test of McFadden (1990) on the remaining samples showed that before bedding correction the remanence directions and bedding dips were correlated at 99% confidence level, whereas after the corrections there was no correlation. This demonstrates that the remanence of the siliceous sinter predates the deformation. If, as seems likely, the sinter strata were deformed while still plastic, this remanence closely dates the primary deposition. The hard and intermediate components give, respectively, the pole estimates 160.4° E, 44.6° S (A_{95} = 21.7°) and 157.8° E, 35.3° S (A_{95} = 20.1°). These estimates are approximate: the geometry of deformation is complex and, therefore, the simple rotational corrections used here to restore the laminae to palaeohorizontal may not be completely accurate.

The low unblocking temperature component in the fine-grained sinter is again interpreted as a viscous remanence produced by recent fields. Both the intermediate and high unblocking temperature components appear to be primary because they pass the fold test. The existence

of two primary magnetizations may be explained by the presence in the sinter of two types of magnetic carrier, fine-grained specularite and hematite pigment, each recording independently, and slightly differently, the direction of the contemporaneous geomagnetic field.

DISCUSSION

Measurements obtained in this study have yielded two poles of sufficiently high precision to be compared with the Australian pole path: one MPD(1) calculated from the intermediate component in the diamictite falls on the Permo-Carboniferous trajectory; the other MPG(H) from the hard component of the granitic breccia falls close to the same trajectory (Fig. 10). These poles indicate two distinct periods of magnetization, the diamictite pole representing the younger event. The Permo-Carboniferous path for Australia is not sufficiently well calibrated and some of its parts are open to alternative interpretation (Klootwijk 1988), so that narrower age assignments are not possible within this period. However, work in progress suggests that the two ages are likely to be mid-Carboniferous and Late Carboniferous/Early Permian (Klootwijk pers. comm. 1990). The poles are far removed from the Cambro-Ordovician (i.e. Delamerian) part of the Australian pole path (Klootwijk 1988; Li *et al.* 1990).

Other poles of this study, from the fine-grained siliceous sinter, hematitic breccia and granitic breccia (intermediate component) are poorly defined. These poles are consistent with the Permo-Carboniferous path, but the 95% confidence circles intersect younger path segments as well (Fig. 10). Consistently positive inclinations of remanence suggest that the ages may be Permo-Carboniferous, because this is the only period that had a prolonged reversed polarity (the Kiaman chron). Support for a minimum Permo-Carboniferous age for the sinter is provided by the presence of only one component of remanence in the diamictite (besides the random component in the cobbles): if the sinter had been deposited after the Permo-Carboniferous, we would expect the diamictite to contain a younger magnetic overprint corresponding to the sinter-forming hydrothermal event. Also, no major post-Palaeozoic tectonism is known for the region other than the late medial Miocene to Pleistocene uplift of the Flinders Ranges (Callen 1977), and the pole positions found here are not consistent with these periods.

A primary depositional origin of the magnetization in the fine-grained siliceous sinter at Mt Gee is demonstrated by a definitive fold test. The sinter thus records major geothermal activity of probable Permo-Carboniferous age. The remanent magnetizations of the diamictite (intermediate component) and at least one of the granitic breccia components are due to overprinting. The remanences of the hematitic breccia and the other granitic

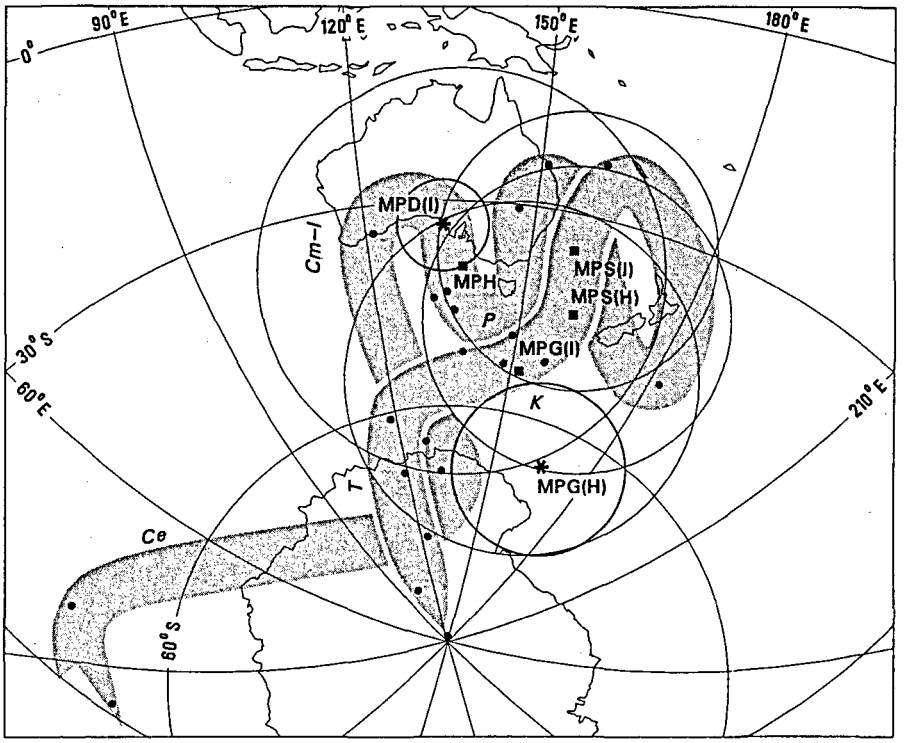


Fig. 10 Carboniferous–Tertiary apparent polar wander path for Australia. The Permo-Carboniferous part of the path is after Li *et al.* (1989), the Triassic to Early Cretaceous part is after Embleton (1981), and the Late Cretaceous to Holocene part is after Idnurm (1985). Small symbols denote calibration poles as defined in the respective references. Large symbols with 95% confidence circles summarize the results from this study.

breccia component are also overprints if the monazite ages of the former are correct; otherwise they could be primary magnetizations.

As noted earlier, we found no evidence in the diamictite of the high unblocking temperature component of Permo-Carboniferous age that is dominant in the enclosing granitic breccia. This suggests that the diamictite formed after the breccia had acquired its high unblocking temperature component, and that it is, therefore, Permo-Carboniferous. The diamictite may well have originated as a Permian tillite (rather than Sturtian; cf. Youles 1975).

We argue that, if the magnetizations of the granitic (hard component) and hematitic breccia are overprints, then they are probably of chemical rather than purely thermal origin. As noted, the high-temperature component in the granitic and hematitic breccia units is reduced only slightly when the samples are heated to temperatures close to the Curie Point of hematite. Theoretical studies (Dunlop & Buchan 1977; Walton 1980) on single-domain hematite suggest that to produce a natural thermal overprint of such high stability would require exposure to temperatures above 600° C for hundreds of Ma (Pullaiah *et al.* 1975; Middleton & Schmidt 1982). Allowing for uncertainties in the assumptions made in the theoretical calculations, Pullaiah *et al.* (1975) suggested more conservatively that such overprints would require upper greenschist or probably lower amphibolite facies regional metamorphism. In the absence of any petrologic evidence of metamorphism or recrystallization (Drexel & Major 1987) a purely thermal overprint thus appears an unlikely explanation for this hard magnetization. A secondary chemical origin for the remanence, imposed by hydrothermal activity, is far more likely in light of the extensive potash metasomatism, reddening and quartz ± hematite veining (even though some of the K-feldspar alteration may pre-date brecciation; Drexel 1980).

GEOLOGICAL INTERPRETATION

The nature of this inferred chemical-hydrothermal overprint, and particularly its relation to uranium mineralization in the hematitic breccia, remains open to speculation. An interpretation accounting for both the pre-Silurian U–Pb monazite ages obtained from the hematitic breccia (Pidgeon 1979) and the Permo-Carboniferous magnetization ages determined in the present study would have to invoke hydrothermal activity over an extended period.

If the monazite age is accepted, the following two-stage scenario may be envisaged in analogy with other ironstone-hosted Cu–U–Au–REE associations, as discussed by Wall (1990) and Wall and Valenta (1990). In a first stage, magmatic-hydrothermal fluids (as recorded by high-temperature fluid inclusions in monazite; Collins 1977, quoted in Drexel & Major 1990) of inferred high Fe^{II} content produce the original iron-rich breccia, precipitating at least some of the iron as magnetite and pyrite. Such a partly reduced ironstone precursor of the breccia is consistent with (1) common pseudomorphs of hematite after magnetite; (2) the constant near-zero $\delta^{34}S$ composition of the pyrite which indicates moderately reducing conditions during precipitation, and (3) the carbon and oxygen stable isotopes of carbonate minerals which suggest a high-temperature rock-buffered fluid source (Lambert *et al.* 1982). In a second stage, U (Cu) mineralization is localized by chemical reaction of oxidized U (Cu)-bearing fluids with pre-existing magnetite-bearing ironstone. This is suggested by unusually low U-contents of the early monazite (Pidgeon 1979), and a paragenetically late occurrence of uraninite + low-iron copper sulphide minerals + hematite (Drexel & Major 1987, 1990). This second-stage hydrothermal oxidation event may have occurred during the Permo-Carboniferous, when secondary magnetizations were imposed on the breccia and diamictite, and when the sinter was deposited at points of fluid discharge to the surface. Such a two-stage hydrothermal evolution is consistent with cross-cutting of granitic breccia by vuggy vein quartz of the Mt Gee type, and with much lower fluid inclusion homogenization temperatures in the latter (cf. Youles 1986). Alternatively, if the geochronological data of Pidgeon (1979) were discarded as an artefact of lead inheritance in U-poor monazite, the entire hydrothermal history including the formation of the ironstone could be Permo-Carboniferous. The Permo-Carboniferous hydrothermal activity was probably triggered by tectonism, and fluid circulation may have been driven by heat related to rapid uplift of radiogenically heated basement rocks or, perhaps, to shallow granitic magmatism.

In a wider geographic context, the Permo-Carboniferous was a period of major tectonism throughout eastern and central Australia. The Kanimblan Orogeny in the Tasman Fold Belt was associated with widespread granitic magmatism. Two samples of granite from the basement beneath the Moomba Gas Field in the Great Artesian Basin, some 250 km north-northeast of Mt Painter, have recently been dated by U–Pb zircon method as Carboniferous (R. B. Flint pers. comm. 1991). The Alice Springs Orogeny, in central Australia, culminated in the Carboniferous (Armstrong & Stewart 1975; Shaw *et al.* 1984, 1991; Mortimer *et al.* 1987).

Magnetic overprints with steep inclinations have also been found in central Australia and are generally attributed to the Alice Springs Orogeny (McWilliams 1977; Kirschvink 1978; Klootwijk 1980; Whiting 1986; Li *et al.* 1989). Li *et al.* (1989) summarized these overprint data and found that bedding-corrected poles form a scattered group centred just south of Australia (Fig. 11). Some of the scatter may be due to structural over-correction, except for pole EL(B) for which a syndeformational origin of remanence has been taken into account. The Mt Painter diamictite pole lies on the same part of the pole path as the centre of the overprint group and particularly the pole EL(B), which suggests that the diamictite may have been magnetized at the time of the overprinting in central Australia. The evidence from Mt Painter thus confirms the suggestion (Powell 1984) that the stresses causing the Alice Springs Orogeny and the Tasman Fold Belt during the Permo-Carboniferous have led to far more widespread tectonism in central and eastern Australia than previously documented.

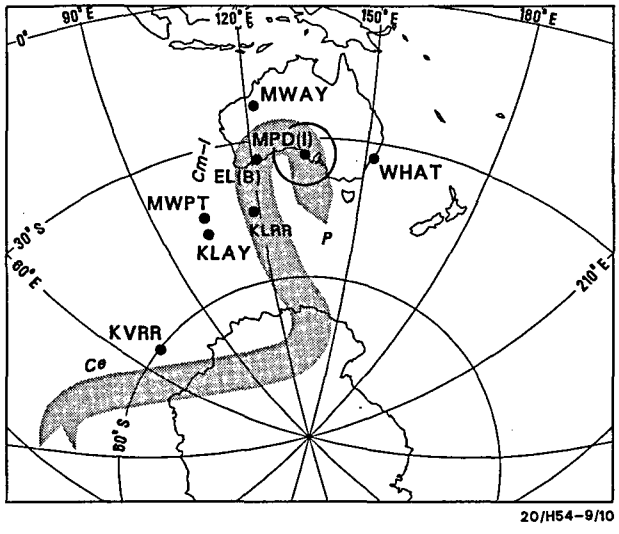


Fig. 11 Diamictite pole MPD(I) for intermediate temperature component compared with overprint poles from central Australia. The acronyms for the latter are as in Li *et al.* (1989). Pole WHAT has not been corrected for bedding dip; pole EL(B) is structurally corrected to allow for syn-deformational overprinting; the other poles are bedding-corrected without allowance for syn-deformational overprinting. The directional scatter among the overprint poles may be due to structural over-corrections.

Finally, we note that there is no evidence in this study for prolonged Tertiary chemical weathering in any of the breccia units (cf. Blissett, in Coats & Blissett 1971), which is characterized by mixed polarities elsewhere in Australia (e.g. Schmidt & Embleton 1976; Idnurm & Senior 1978).

CONCLUSIONS

The following sequence of events during the Permo-Carboniferous is inferred from the palaeomagnetic data.

(i) An intense remagnetization of the breccia which destroyed any previous stable remanence or, alternatively, the formation of the granitic breccia and ironstone (hematitic breccia).

(ii) A second magnetization event in the granitic breccia which partly overprinted the earlier remanence.

(iii) Formation of diamictite bodies within the breccia some time after the first magnetization. Randomly directed magnetizations of the rounded cobbles were overprinted after the cobbles had been deposited, possibly during the younger remagnetization event.

(iv) Deposition of the Mt Gee Sinter by hydrothermal processes that probably caused the younger magnetization.

Palaeomagnetic results indicate major near-surface hydrothermal activity in the Mt Painter region during the Permo-Carboniferous, which included sinter deposition from hot springs (the Mt Gee Sinter), and possibly ironstone formation and uranium mineralization as well. However, if earlier radiometric data are correct, the deposition of ironstone and the Radium Ridge Breccias could have considerably pre-dated sinter-forming activity. The hydrothermal activity in the Permo-Carboniferous produced a hard secondary magnetization in the breccia, probably as a consequence of chemical overprinting under oxidizing conditions. This would indicate that uranium mineralization hosted by the hematitic breccia resulted from interaction, during the Permo-Carboniferous, of oxidized fluids with pre-existing magnetite in the breccia. Nevertheless uranium mineralization as a primary part of the ironstone-forming process cannot be excluded.

The study suggests that tectonic activity associated with the Alice Springs Orogeny was more extensive than previously known, and may well have triggered ore-forming epithermal activity elsewhere in central Australia. This study also exemplifies that palaeomagnetic data, if combined with detailed geological observation, can usefully complement radiogenic isotope studies in dating hydrothermal ore-forming events.

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